

The incorrect h tag could cause the machine freeze due to excessive memory use #143

Closed

vokiput opened on Aug 27, 2024

markdownify in `./venv/lib/python3.8/site-packages (0.13.1)`

The issue is found with atheris library

The code to reproduce the issue:

```
markdownify("<html><body><h5555555555>My First Heading</h5555555555><p>My first paragraph.</p></body></html>")
```

My machine had frozen. Ubuntu 20.04. 16 GB. The memory usage went to 100% in 2-3 seconds. The only way to fix it is to turn it off/on.

The only valid cases are h1 - h6. We should ignore everything else.

It could be an edge case but it could be possible to feed the string in the example into a server to cause resource exhaustion.

Related cases are (will be fixed if we fix the original issue)

```
import sys
markdownify(f"<h5{sys.maxsize // 10}>")
Traceback (most recent call last):
  File "/home/redacted/code/other/atheris/pythonProject/test_unit_009-1.py", line 22, in
<module>
    markdownify(f"<h5{sys.maxsize // 10}>")
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 433, in markdownify
    return MarkdownConverter(**options).convert(html)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 105, in convert
    return self.convert_soup(soup)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 108, in convert_soup
    return self.process_tag(soup, convert_as_inline=False, children_only=True)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 151, in process_tag
    text += self.process_tag(el, convert_children_as_inline)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 156, in process_tag
    text = convert_fn(node, text, convert_as_inline)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 188, in convert_tag
```

```
        return self.convert_hn(n, el, text, convert_as_inline)
    File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 283, in convert_hn
        hashes = '#' * n
MemoryError
```

and

```
import sys
markdownify(f"<h{sys.maxsize + 1}>")
Traceback (most recent call last):
  File "/home/redacted/code/other/atheris/pythonProject/test_unit_009-1.py", line 15, in
<module>
    markdownify(f"<h{sys.maxsize + 1}>")
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 433, in markdownify
    return MarkdownConverter(**options).convert(html)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 105, in convert
    return self.convert_soup(soup)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 108, in convert_soup
    return self.process_tag(soup, convert_as_inline=False, children_only=True)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 151, in process_tag
    text += self.process_tag(el, convert_children_as_inline)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 156, in process_tag
    text = convert_fn(node, text, convert_as_inline)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 188, in convert_tag
    return self.convert_hn(n, el, text, convert_as_inline)
  File "/home/redacted/code/other/atheris/pythonProject/.venv/lib/python3.8/site-
packages/markdownify/__init__.py", line 283, in convert_hn
    hashes = '#' * n
OverflowError: cannot fit 'int' into an index-sized integer
```

  **AlexVonB** closed this as completed in [9595618](#) on Nov 25, 2024

 AlexVonB on Nov 25, 2024

Collaborator ...

Good catch, thank you!

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  **AlexVonB** added a commit that references this issue on Nov 25, 2024

prevent very large headline prefixes ...

 [9595618](#)

  **Wuhall** added a commit that references this issue on May 21, 2025

prevent very large headline prefixes ...

 [0afab1e](#)

  **alheiveea** mentioned this on Jul 9, 2025

 High colspan values can cause denial of service on servers or machine to freeze #231

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